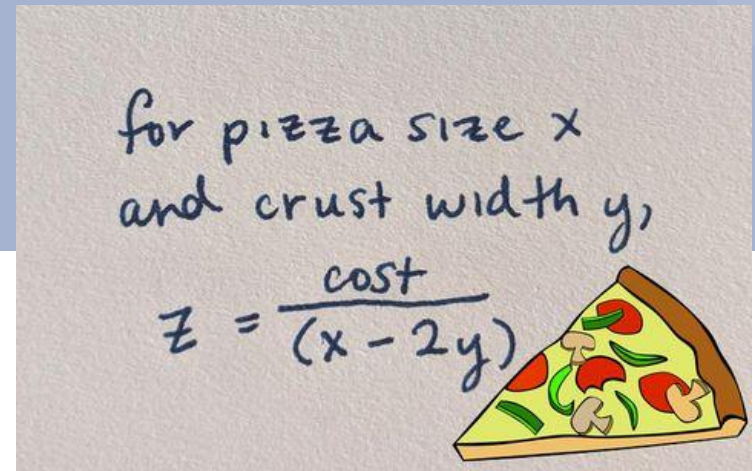


CS2300: File Structures and Introduction to Database Systems

Lecture 7: Relational Algebra

San Yeung



relational algebra

set operations

relational database specific operations

set functions



set union



selection

sum



set intersection

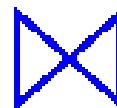


projection

avg



set difference

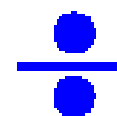


join

count



cartesian product



set division

any

max

min

Introduction

- **Relational Algebra** is a **formal language** designed for the relational model to represent the basic set of **operations** for data retrieval requests
- The SQL language is primarily based on concepts from *relational calculus* and relational algebra
- **Relational Algebra Expression:**
 - A sequence of relational algebra operations whose result is a "new" relation that represents the result of a database query
- **Why is relational algebra important?**
 1. RA provides a formal foundation for relational model operations
 2. RA is used as a basis for implementing and optimizing queries in the query processing and optimization modules that are integral parts of RDBMSs
 3. RA provides a formal foundation for the SQL standard query language

Relational Algebra

- Relational algebra has a collection of operators on *relations*
- Operators may be *unary* or *binary*
- The output of an operator is a relation
 - Another way to say this is that the algebra is “closed”
 - Therefore, operators can be composed one on another

Basic Operators Overview

- Relational Algebra has 5 basic operators:
 - **Project (π)**
 - **Select (σ)**
 - **Union (U)**
 - **Set difference (-)**
 - **Cartesian product (X)**
- Other operators can be defined using the basic ones: Intersection, Join, Division ...
- A useful syntactic operator: Rename



Unary Relational Operations:

SELECT AND PROJECT

The SELECT Operation

- This operation is used to choose a *subset* of the tuples from a relation that satisfies a **selection condition**
- Notation: $\sigma_{\langle \text{selection condition} \rangle}(R)$
- R can be:
 - a simple database relation, or,
 - a *relational algebra expression* whose result is a relation
- The *selection condition*:
 - a comparison between an attribute and a constant value/another attribute
 - *comparison operators*: $<$, $\leq$, $>$, $\geq$, $=$, $\neq$
 - the condition clause can be connected by Boolean operators *and*, *or*, and *not*

Types of Comparisons

- We can use any of the operators:

$<$, $\leq$, $>$, $\geq$, $=$, $\neq$

- When comparing an attribute with a string, the string is written in single quotes

$$\sigma_{tname='cohen'}(T)$$

Important! Conditions are evaluated a single tuple at a time.

EMPLOYEE

Fname	Minit	Lname	<u>Ssn</u>	Bdate	Address	Sex	Salary	Super_ssn	Dno
John	B	Smith	123456789	1965-01-09	731 Fondren, Houston, TX	M	30000	333445555	5
Franklin	T	Wong	333445555	1955-12-08	638 Voss, Houston, TX	M	40000	888665555	5
Alicia	J	Zelaya	999887777	1968-01-19	3321 Castle, Spring, TX	F	25000	987654321	4
Jennifer	S	Wallace	987654321	1941-06-20	291 Berry, Bellaire, TX	F	43000	888665555	4
Ramesh	K	Narayan	666884444	1962-09-15	975 Fire Oak, Humble, TX	M	38000	333445555	5
Joyce	A	English	453453453	1972-07-31	5631 Rice, Houston, TX	F	25000	333445555	5
Ahmad	V	Jabbar	987987987	1969-03-29	980 Dallas, Houston, TX	M	25000	987654321	4
James	E	Borg	888665555	1937-11-10	450 Stone, Houston, TX	M	55000	NULL	1

$\sigma_{(Dno=4 \text{ AND } Salary>25000) \text{ OR } (Dno=5 \text{ AND } Salary>30000)}(EMPLOYEE)$

Examples

- The selection operation is applied to each tuple *individually*
- The number of tuples in the resulting relation is always _____ to the number of tuples in R
 - and the minimum?
- The fraction of tuples selected is referred to as the **selectivity** of the selection condition

Examples

$\sigma_{(Dno=4 \text{ AND } Salary > 25000) \text{ OR } (Dno=5 \text{ AND } Salary > 30000)}(EMPLOYEE)$

Fname	Minit	Lname	<u>Ssn</u>	Bdate	Address	Sex	Salary	Super_ssn	Dno
Franklin	T	Wong	333445555	1955-12-08	638 Voss, Houston, TX	M	40000	888665555	5
Jennifer	S	Wallace	987654321	1941-06-20	291 Berry, Bellaire, TX	F	43000	888665555	4
Ramesh	K	Narayan	666884444	1962-09-15	975 Fire Oak, Humble, TX	M	38000	333445555	5

The PROJECT Operation

- This operation is used to choose a *subset* of the *columns* from a relation based on the attributes that we are interested in
- Notation: $\pi_{\langle \text{attribute list} \rangle}(R)$
- The result of this operation has only the attributes specified in *<attribute list>* in the *same order* as they appear in the list

$\pi_{\text{Lname, Fname, Salary}}(\text{EMPLOYEE})$

Lname	Fname	Salary
Smith	John	30000
Wong	Franklin	40000
Zelaya	Alicia	25000
Wallace	Jennifer	43000
Narayan	Ramesh	38000
English	Joyce	25000
Jabbar	Ahmad	25000
Borg	James	55000

EMPLOYEE

Fname	Minit	Lname	<u>Ssn</u>	Bdate	Address	Sex	Salary	Super_ssn	Dno
John	B	Smith	123456789	1965-01-09	731 Fondren, Houston, TX	M	30000	333445555	5
Franklin	T	Wong	333445555	1955-12-08	638 Voss, Houston, TX	M	40000	888665555	5
Alicia	J	Zelaya	999887777	1968-01-19	3321 Castle, Spring, TX	F	25000	987654321	4
Jennifer	S	Wallace	987654321	1941-06-20	291 Berry, Bellaire, TX	F	43000	888665555	4
Ramesh	K	Narayan	666884444	1962-09-15	975 Fire Oak, Humble, TX	M	38000	333445555	5
Joyce	A	English	453453453	1972-07-31	5631 Rice, Houston, TX	F	25000	333445555	5
Ahmad	V	Jabbar	987987987	1969-03-29	980 Dallas, Houston, TX	M	25000	987654321	4
James	E	Borg	888665555	1937-11-10	450 Stone, Houston, TX	M	55000	NULL	1

$\pi_{\text{Sex, Salary}}(\text{EMPLOYEE})$

Examples

- **Duplicate Elimination**
 - **duplicate tuples** are likely to occur if nonkey attributes are included in <attribute list>
 - a PROJECTION removes any duplicate tuples from the *result* to maintain a *valid* relation
- The number of tuples in the resulting relation is always _____ to the number of tuples in *R*
 - and the minimum?

Example

- How would you find the names of the third year students?

S

<u>sid</u>	sname	year
240	white	3
202	jones	3
450	adams	1

?

sname
white
Jones



Sequences of Operations and the RENAME Operation

- A relational algebra operation can be represented as a single expression (**in-line expression**)
- Example)
 - $\pi_{FNAME, LNAME, SALARY}(\sigma_{DNO=5}(EMPLOYEE))$
- Alternatively, a sequence of operations can be used by giving a name to each intermediate **relation** and using the *assignment operation* ($\leftarrow$):
 - $DEP5_EMPS \leftarrow \sigma_{DNO=5}(EMPLOYEE)$
 - $RESULT \leftarrow \pi_{FNAME, LNAME, SALARY}(DEP5_EMPS)$
 - $R(\text{First_name}, \text{Last_name}, \text{Salary}) \leftarrow RESULT$

EMPLOYEE

Fname	Minit	Lname	<u>Ssn</u>	Bdate	Address	Sex	Salary	Super_ssn	Dno
John	B	Smith	123456789	1965-01-09	731 Fondren, Houston, TX	M	30000	333445555	5
Franklin	T	Wong	333445555	1955-12-08	638 Voss, Houston, TX	M	40000	888665555	5
Alicia	J	Zelaya	999887777	1968-01-19	3321 Castle, Spring, TX	F	25000	987654321	4
Jennifer	S	Wallace	987654321	1941-06-20	291 Berry, Bellaire, TX	F	43000	888665555	4
Ramesh	K	Narayan	666884444	1962-09-15	975 Fire Oak, Humble, TX	M	38000	333445555	5
Joyce	A	English	453453453	1972-07-31	5631 Rice, Houston, TX	F	25000	333445555	5
Ahmad	V	Jabbar	987987987	1969-03-29	980 Dallas, Houston, TX	M	25000	987654321	4
James	E	Borg	888665555	1937-11-10	450 Stone, Houston, TX	M	55000	NULL	1

The RENAME Operation

- This operation is used to **rename** the relation name or the attribute names, or both
- Notation: $\rho_{S(B_1, B_2, \dots, B_n)}(R)$ or $\rho_S(R)$ or $\rho_{(B_1, B_2, \dots, B_n)}(R)$
- S – the new relation name
- $B_1, B_2, \dots, B_n$ – the new attribute names in *order*
- This can be useful in connection with more complex operations such as relational JOIN

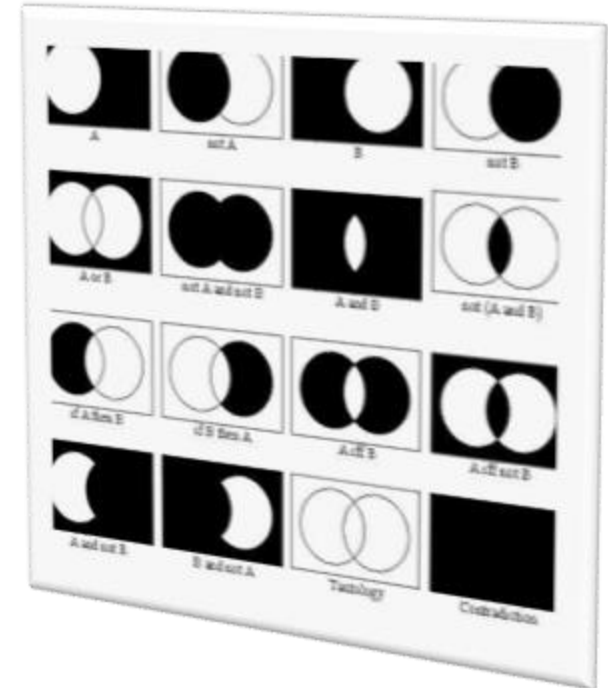
$DEP5_EMPS \leftarrow \sigma_{DNO=5}(EMPLOYEE)$

$RESULT \leftarrow \pi_{FNAME, LNAME, SALARY}(DEP5_EMPS)$

$\rho_{R(First_name, Last_name, Salary)}(RESULT)$

R

First_name	Last_name	Salary
John	Smith	30000
Franklin	Wong	40000
Ramesh	Narayan	38000
Joyce	English	25000



Relational Operations from Set Theory:

UNION, INTERSECTION, MINUS, AND CARTESIAN PRODUCT

The Union Operation

- Binary operator; Written as $R \cup S$
- Union can only be performed between *compatible* relations
 - *same number* of attributes
 - *corresponding* attributes have the same type
- Result contains all tuples in at least one of the relations
- The attribute names from R are used as the schema of the result
- Duplicate tuples are eliminated

Example

S

<u>sid</u>	sname	year
240	white	3
202	jones	3
450	adams	1

G

<u>sid</u>	sname	year
701	katz	1
202	jones	3
820	sapir	2

$S \cup G$

<u>sid</u>	sname	year
240	white	3
202	jones	3
701	katz	1
820	sapir	2
450	adams	1

Example

 S

<u>sid</u>	sname	year
240	white	3
202	jones	3

 G

<u>ID</u>	Name	year
202	jones	3

 $S \cup G ?$

<u>sid</u>	sname	year
240	white	3
202	jones	3

Intersection

- Binary operator
- Written as $R \cap S$
- Performed between **compatible** relations
- Result contains the tuples that appear in R and S

Example

S

sid	sname	year
240	white	3
202	jones	3
450	adams	1

G

id	sname	year
701	katz	1
202	jones	3
820	sapir	2

$S \cap G$

What is in the
result?

sid	sname	year
202	jones	3

The Set Difference Operation

- Binary operator; written as $R - S$
 - also called MINUS or EXCEPT
- Can only be performed between relations that are *compatible*
- Result contains the tuples from R, but not in S
- Schema attributes of result are those of R

Example

S

<u>sid</u>	sname	year
240	white	3
202	jones	3
450	adams	1

G

<u>ID</u>	sname	year
701	katz	1
202	jones	3
820	sapir	2

S – G

<u>sid</u>	sname	year
240	white	3
450	adams	1

What about G-S??

Exercise

1) Write a relational algebra expression to retrieve the students (sid) who did not take the “db” course using the set difference operator.

R

tid	sid	course
20	202	os
10	450	calculus
20	202	db
20	240	db

Some properties of UNION, INTERSECT, and DIFFERENCE

- Notice that both union and intersection are *commutative* operations; that is
 - $R \cup S = S \cup R$, and $R \cap S = S \cap R$
- Both union and intersection can be treated as **n-ary operations** applicable to any number of relations as both are *associative* operations; that is
 - $R \cup (S \cup T) = (R \cup S) \cup T$
 - $(R \cap S) \cap T = R \cap (S \cap T)$
- The minus operation is not commutative; that is, in general
 - $R - S \neq S - R$

The CARTESIAN PRODUCT Operation

- This *binary* operation is used to **combine** tuples from two relations in a combinatorial fashion without requiring them to be compatible
- Notation: **$R \times S$**
- It produces a new relation by combining **every** tuple from one relation with **every** tuple from the other relation

R

First_name	Last_name	Salary
John	Smith	30000
Franklin	Wong	40000
Ramesh	Narayan	38000
Joyce	English	25000

 $\times$

R

First_name	Last_name	Salary
John	Smith	30000
Franklin	Wong	40000
Ramesh	Narayan	38000
Joyce	English	25000

?

Cartesian Product (Cross Product)

- Binary Operator;
- Denoted by
 - $R(A_1, A_2, \dots, A_n) \times S(B_1, B_2, \dots, B_m)$
- Result is a relation Q with degree $n + m$ **attributes**:
 - $Q(A_1, A_2, \dots, A_n, B_1, B_2, \dots, B_m)$, in that order.
- Do R and S **need to be compatible?**

How to find the course taught by each teacher?

R

<u>tid</u>	<u>sid</u>	<u>course</u>
20	202	os
10	450	calculus
20	202	db
20	240	db

T

<u>tid</u>	tname	dept
10	Cohen	math
20	Levy	cs

R = Studies

T = Teachers

Generally, CROSS PRODUCT is not a meaningful operation unless? What additional queries can we apply?

R.tid

 $R \times T$

T.tid

(tid)	sid	course	(tid)	tname	dept
20	202	os	10	cohen	math
10	450	calculus	10	cohen	math
20	202	db	10	cohen	math
20	240	db	10	cohen	math
20	202	os	20	levy	cs
10	450	calculus	20	levy	cs
20	202	db	20	levy	cs
20	240	db	20	levy	cs

$R \times T$

Both columns have the same name,
result is undefined

(tid)	sid	course	(tid)	tname	dept
20	202	os	10	cohen	math
10	450	calculus	10	cohen	math
20	202	db	10	cohen	math
20	240	db	10	cohen	math
20	202	os	20	levy	cs
10	450	calculus	20	levy	cs
20	202	db	20	levy	cs
20	240	db	20	levy	cs

$\rho_{\text{Course}}(\text{tid1}, \text{sid}, \text{course}, \text{tid2}, \text{tname}, \text{dpmnt})$ (*RXT*)

Course

tid1	sid	course	tid2	tname	dpmnt
20	202	os	10	cohen	math
10	450	calculus	10	cohen	math
20	202	db	10	cohen	math
20	240	db	10	cohen	math
20	202	os	20	levy	cs
10	450	calculus	20	levy	cs
20	202	db	20	levy	cs
20	240	db	20	levy	cs

$\sigma_{tid1=tid2} (\rho_{Course} (tid1, sid, course, tid2, tname, dpmnt) (RXT))$

Course

tid1	sid	course	tid2	tname	dept
20	202	os	10	cohen	math
10	450	calculus	10	cohen	math
20	202	db	10	cohen	math
20	240	db	10	cohen	math
20	202	os	20	levy	cs
10	450	calculus	20	levy	cs
20	202	db	20	levy	cs
20	240	db	20	levy	cs

$\Pi_{\text{course, tid2, tname}} (\sigma_{\text{tid1=tid2}} (\rho_{\text{Course}} (\text{tid1, sid, course, tid2, tname, dpmnt})) (RXT))$

Course

tid1	sid	course	tid2	tname	dept
10	450	calculus	10	cohen	math
20	202	os	20	levy	cs
20	202	db	20	levy	cs
20	240	db	20	levy	cs

$\Pi_{\text{course, tid2, tname}} (\sigma_{\text{tid1=tid2}} (\rho_{\text{Course}} (\text{tid1, sid, course, tid2, tname, dpmnt})) (RXT))$

Result

course	tid2	tname
calculus	10	cohen
os	20	levy
db	20	levy

Question

R

<u>tid</u>	<u>sid</u>	<u>course</u>
20	202	os
10	450	calculus
20	202	db
20	240	db

Find the names of the courses taught by Levy

T

<u>tid</u>	tname	dept
10	cohen	math
20	levy	cs

$\rho_{\text{Course}}(\text{tid1}, \text{sid}, \text{course}, \text{tid2}, \text{tname}, \text{dpmnt})$ (*RXT*)

Course

tid1	sid	course	tid2	tname	dpmnt
20	202	os	10	cohen	math
10	450	calculus	10	cohen	math
20	202	db	10	cohen	math
20	240	db	10	cohen	math
20	202	os	20	levy	cs
10	450	calculus	20	levy	cs
20	202	db	20	levy	cs
20	240	db	20	levy	cs

$\sigma_{tid1=tid2} (\rho_{Course} (tid1, sid, course, tid2, tname, dpmnt) (RXT))$

Course

tid1	sid	course	tid2	tname	dept
20	202	os	10	cohen	math
10	450	calculus	10	cohen	math
20	202	db	10	cohen	math
20	240	db	10	cohen	math
20	202	os	20	levy	cs
10	450	calculus	20	levy	cs
20	202	db	20	levy	cs
20	240	db	20	levy	cs

$\sigma_{tid1=tid2 \text{ AND } tname = 'levy'}$ ($\rho_{Course (tid1,sid,course,tid2,tname,dpmnt)}(RXT)$)

Course

tid1	sid	course	tid2	tname	dept
10	450	calculus	10	cohen	math
20	202	os	20	levy	cs
20	202	db	20	levy	cs
20	240	db	20	levy	cs

$$\Pi_{\text{course}}(\sigma_{\text{tid1}=\text{tid2 AND tname} = \text{'levy'}}(\rho_{\text{Course}}(\text{tid1}, \text{sid}, \text{course}, \text{tid2}, \text{tname}, \text{dpmnt})(RXT)))$$

Course

tid1	sid	course	tid2	tname	dept
20	202	os	20	levy	cs
20	202	db	20	levy	cs
20	240	db	20	levy	cs

$\Pi_{\text{course}}(\sigma_{\text{tid1}=\text{tid2 AND tname} = \text{'levy'}}(\rho_{\text{Course}}(\text{tid1}, \text{sid}, \text{course}, \text{tid2}, \text{tname}, \text{dpmnt})(RXT)))$

Result

course
os
db

Examples

- The CARTESIAN PRODUCT operation applied by itself is generally meaningless
- It must be accompanied by other pairing operations
- For example:
 - *How to retrieve the list of names of each female employee's dependents?*

EMPLOYEE

Fname	Minit	Lname	Ssn	Bdate	Address	Sex	Salary	Super_ssn	Dno
John	B	Smith	123456789	1965-01-09	731 Fondren, Houston, TX	M	30000	333445555	5
Franklin	T	Wong	333445555	1955-12-08	638 Voss, Houston, TX	M	40000	888665555	5
Alicia	J	Zelaya	999887777	1968-01-19	3321 Castle, Spring, TX	F	25000	987654321	4
Jennifer	S	Wallace	987654321	1941-06-20	291 Berry, Bellaire, TX	F	43000	888665555	4
Ramesh	K	Narayan	666884444	1962-09-15	975 Fire Oak, Humble, TX	M	38000	333445555	5
Joyce	A	English	453453453	1972-07-31	5631 Rice, Houston, TX	F	25000	333445555	5
Ahmad	V	Jabbar	987987987	1969-03-29	980 Dallas, Houston, TX	M	25000	987654321	4
James	E	Borg	888665555	1937-11-10	450 Stone, Houston, TX	M	55000	NULL	1

DEPARTMENT

Dname	Dnumber	Mgr_ssn	Mgr_start_date
Research	5	333445555	1988-05-22
Administration	4	987654321	1995-01-01
Headquarters	1	888665555	1981-06-19

DEPT_LOCATIONS

Dnumber	Dlocation
1	Houston
4	Stafford
5	Bellaire
5	Sugarland
5	Houston

WORKS_ON

Essn	Pno	Hours
123456789	1	32.5
123456789	2	7.5
666884444	3	40.0
453453453	1	20.0
453453453	2	20.0
333445555	2	10.0
333445555	3	10.0
333445555	10	10.0
333445555	20	10.0
999887777	30	30.0
999887777	10	10.0
987987987	10	35.0
987987987	30	5.0
987654321	30	20.0
987654321	20	15.0
888665555	20	NULL

PROJECT

Pname	Pnumber	Plocation	Dnum
ProductX	1	Bellaire	5
ProductY	2	Sugarland	5
ProductZ	3	Houston	5
Computerization	10	Stafford	4
Reorganization	20	Houston	1
Newbenefits	30	Stafford	4

DEPENDENT

Essn	Dependent_name	Sex	Bdate	Relationship
333445555	Alice	F	1986-04-05	Daughter
333445555	Theodore	M	1983-10-25	Son
333445555	Joy	F	1958-05-03	Spouse
987654321	Abner	M	1942-02-28	Spouse
123456789	Michael	M	1988-01-04	Son
123456789	Alice	F	1988-12-30	Daughter
123456789	Elizabeth	F	1967-05-05	Spouse

Examples

FEMALE_EMPS

Fname	Minit	Lname	Ssn	Bdate	Address	Sex	Salary	Super_ssn	Dno
Alicia	J	Zelaya	999887777	1968-07-19	3321 Castle, Spring, TX	F	25000	987654321	4
Jennifer	S	Wallace	987654321	1941-06-20	291 Berry, Bellaire, TX	F	43000	888665555	4
Joyce	A	English	453453453	1972-07-31	5631 Rice, Houston, TX	F	25000	333445555	5

EMPNAMES

Fname	Lname	Ssn
Alicia	Zelaya	999887777
Jennifer	Wallace	987654321
Joyce	English	453453453

EMP_DEPENDENTS

Fname	Lname	Ssn	Essn	Dependent_name	Sex	Bdate	...
Alicia	Zelaya	999887777	333445555	Alice	F	1986-04-05	...
Alicia	Zelaya	999887777	333445555	Theodore	M	1983-10-25	...
Alicia	Zelaya	999887777	333445555	Joy	F	1958-05-03	...
Alicia	Zelaya	999887777	987654321	Abner	M	1942-02-28	...
Alicia	Zelaya	999887777	123456789	Michael	M	1988-01-04	...
Alicia	Zelaya	999887777	123456789	Alice	F	1988-12-30	...
Alicia	Zelaya	999887777	123456789	Elizabeth	F	1967-05-05	...
Jennifer	Wallace	987654321	333445555	Alice	F	1986-04-05	...
Jennifer	Wallace	987654321	333445555	Theodore	M	1983-10-25	...
Jennifer	Wallace	987654321	333445555	Joy	F	1958-05-03	...
Jennifer	Wallace	987654321	987654321	Abner	M	1942-02-28	...
Jennifer	Wallace	987654321	123456789	Michael	M	1988-01-04	...
Jennifer	Wallace	987654321	123456789	Alice	F	1988-12-30	...
Jennifer	Wallace	987654321	123456789	Elizabeth	F	1967-05-05	...
Joyce	English	453453453	333445555	Alice	F	1986-04-05	...
Joyce	English	453453453	333445555	Theodore	M	1983-10-25	...
Joyce	English	453453453	333445555	Joy	F	1958-05-03	...
Joyce	English	453453453	987654321	Abner	M	1942-02-28	...
Joyce	English	453453453	123456789	Michael	M	1988-01-04	...
Joyce	English	453453453	123456789	Alice	F	1988-12-30	...
Joyce	English	453453453	123456789	Elizabeth	F	1967-05-05	...

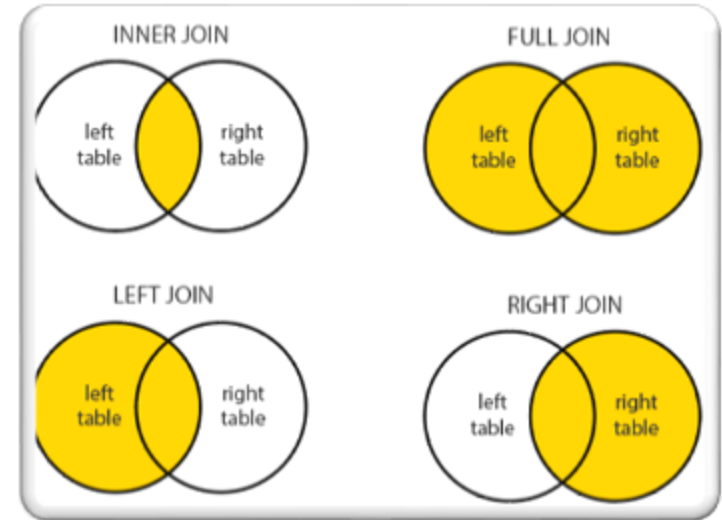
$FEMALE_EMPS \leftarrow \sigma_{Sex='F'}(EMPLOYEE)$
 $EMPNAMES \leftarrow \pi_{Fname, Lname, Ssn}(FEMALE_EMPS)$
 $EMP_DEPENDENTS \leftarrow EMPNAMES \times DEPENDENT$
 $ACTUAL_DEPENDENTS \leftarrow \sigma_{Ssn=Essn}(EMP_DEPENDENTS)$
 $RESULT \leftarrow \pi_{Fname, Lname, Dependent_name}(ACTUAL_DEPENDENTS)$

ACTUAL_DEPENDENTS

Fname	Lname	Ssn	Essn	Dependent_name	Sex	Bdate	...
Jennifer	Wallace	987654321	987654321	Abner	M	1942-02-28	...

RESULT

Fname	Lname	Dependent_name
Jennifer	Wallace	Abner



Binary Relational Operations:

JOIN AND DIVISION

The JOIN Operation

- Join is a “*shortcut*” for writing the combination of selection and cross product

- Notation: $R \bowtie_{\langle \text{join condition} \rangle} S$

- A conditional (theta) join has the format

$$R \bowtie_C S$$

where C is a condition, as in a Select

- Most join conditions involve one or more equality conditions “AND”ed or “OR”ed together
- This expression is equivalent to:

$$\sigma_C(R \times S)$$

Find the students who are taught by
teachers from CS department

R

<u>tid1</u>	<u>sid</u>	<u>course</u>
20	202	os
10	450	calculus
20	202	db
20	240	db

T

<u>tid2</u>	tname	dept
10	Cohen	math
20	Levy	cs

Variations of JOIN: The EQUIJOIN and NATURAL JOIN

- An **EQUIJOIN** is a join where the only comparison operator used in the join conditions is =
- A **NATURAL JOIN** combines two relations based on **common attribute(s)** with the **same name** and removes **duplicate attributes** from the result.

Notation: **R * S**

DEPT_LOCS ← DEPARTMENT * DEPT_LOCATIONS

DEPARTMENT

Dname	Dnumber	Mgr_ssn	Mgr_start_date
Research	5	333445555	1988-05-22
Administration	4	987654321	1995-01-01
Headquarters	1	888665555	1981-06-19

DEPT_LOCS

Dname	Dnumber	Mgr_ssn	Mgr_start_date	Location
Headquarters	1	888665555	1981-06-19	Houston
Administration	4	987654321	1995-01-01	Stafford
Research	5	333445555	1988-05-22	Bellaire
Research	5	333445555	1988-05-22	Sugarland
Research	5	333445555	1988-05-22	Houston

DEPT_LOCATIONS

Dnumber	Dlocation
1	Houston
4	Stafford
5	Bellaire
5	Sugarland
5	Houston

Example Relations

R

<u>tid</u>	<u>sid</u>	<u>course</u>
20	202	os
10	450	calculus
20	202	db
20	240	db

 $R * T = ?$

<u>tid</u>	sid	course	tname	dept
10	450	calculus	cohen	math
20	202	os	levy	cs
20	202	db	levy	cs
20	240	db	levy	cs

T

<u>tid</u>	tname	dept
10	cohen	math
20	levy	cs

A Complete Set of Relational Algebra Operations

- The set of operations including
 - SELECT σ
 - PROJECT π
 - UNION $\cup$
 - DIFFERENCE $\sqsubset$
 - RENAME ρ
 - and CARTESIAN PRODUCT $\times$
 - is called a **complete set** because any other relational algebra expression can be expressed by a combination of these five operations.
- For example:
 - $R \cap S = (R \cup S) - ((R - S) \cup (S - R))$
 - $R \bowtie_{\langle \text{join condition} \rangle} S = \sigma_{\langle \text{join condition} \rangle} (R \times S)$

The DIVISION Operation

- Division is useful to express “for all” queries
- We will use $\div$ to denote division
 - $R(Z) \div S(X)$, where X is a subset of attributes of Z
- Example:
 - Retrieve the names of employees who work on *all* the projects that ‘John Smith’ works on

EMPLOYEE

Fname	Minit	Lname	Ssn	Bdate	Address	Sex	Salary	Super_ssn	Dno
John	B	Smith	123456789	1965-01-09	731 Fondren, Houston, TX	M	30000	333445555	5
Franklin	T	Wong	333445555	1955-12-08	638 Voss, Houston, TX	M	40000	888665555	5
Alicia	J	Zelaya	999887777	1968-01-19	3321 Castle, Spring, TX	F	25000	987654321	4
Jennifer	S	Wallace	987654321	1941-06-20	291 Berry, Bellaire, TX	F	43000	888665555	4
Ramesh	K	Narayan	666884444	1962-09-15	975 Fire Oak, Humble, TX	M	38000	333445555	5
Joyce	A	English	453453453	1972-07-31	5631 Rice, Houston, TX	F	25000	333445555	5
Ahmad	V	Jabbar	987987987	1969-03-29	980 Dallas, Houston, TX	M	25000	987654321	4
James	E	Borg	888665555	1937-11-10	450 Stone, Houston, TX	M	55000	NULL	1

DEPARTMENT

Dname	Dnumber	Mgr_ssn	Mgr_start_date
Research	5	333445555	1988-05-22
Administration	4	987654321	1995-01-01
Headquarters	1	888665555	1981-06-19

DEPT_LOCATIONS

Dnumber	Dlocation
1	Houston
4	Stafford
5	Bellaire
5	Sugarland
5	Houston

WORKS_ON

Essn	Pno	Hours
123456789	1	32.5
123456789	2	7.5
666884444	3	40.0
453453453	1	20.0
453453453	2	20.0
333445555	2	10.0
333445555	3	10.0
333445555	10	10.0
333445555	20	10.0
999887777	30	30.0
999887777	10	10.0
987987987	10	35.0
987987987	30	5.0
987654321	30	20.0
987654321	20	15.0
888665555	20	NULL

PROJECT

Pname	Pnumber	Plocation	Dnum
ProductX	1	Bellaire	5
ProductY	2	Sugarland	5
ProductZ	3	Houston	5
Computerization	10	Stafford	4
Reorganization	20	Houston	1
Newbenefits	30	Stafford	4

DEPENDENT

Essn	Dependent_name	Sex	Bdate	Relationship
333445555	Alice	F	1986-04-05	Daughter
333445555	Theodore	M	1983-10-25	Son
333445555	Joy	F	1958-05-03	Spouse
987654321	Abner	M	1942-02-28	Spouse
123456789	Michael	M	1988-01-04	Son
123456789	Alice	F	1988-12-30	Daughter
123456789	Elizabeth	F	1967-05-05	Spouse

Find suppliers (sno) from A who supply **All parts (pno) from B**

A

<u>sno</u>	<u>pno</u>
S1	P1
S1	P2
S1	P3
S1	P4
S2	P1
S2	P2
S3	P2
S4	P2
S4	P4

B

<u>pno</u>
P2
P4

÷

=

<u>sno</u>
S1
S4

Table 8.1 Operations of Relational Algebra

Table 8.1 Operations of Relational Algebra

OPERATION	PURPOSE	NOTATION
SELECT	Selects all tuples that satisfy the selection condition from a relation R .	$\sigma_{\langle \text{selection condition} \rangle}(R)$
PROJECT	Produces a new relation with only some of the attributes of R , and removes duplicate tuples.	$\pi_{\langle \text{attribute list} \rangle}(R)$
THETA JOIN	Produces all combinations of tuples from R_1 and R_2 that satisfy the join condition.	$R_1 \bowtie_{\langle \text{join condition} \rangle} R_2$
EQUIJOIN	Produces all the combinations of tuples from R_1 and R_2 that satisfy a join condition with only equality comparisons.	$R_1 \bowtie_{\langle \text{join condition} \rangle} R_2$, OR $R_1 \bowtie_{(\langle \text{join attributes 1} \rangle), (\langle \text{join attributes 2} \rangle)} R_2$
NATURAL JOIN	Same as EQUIJOIN except that the join attributes of R_2 are not included in the resulting relation; if the join attributes have the same names, they do not have to be specified at all.	$R_1 \star_{\langle \text{join condition} \rangle} R_2$, OR $R_1 \star_{(\langle \text{join attributes 1} \rangle), (\langle \text{join attributes 2} \rangle)} R_2$ OR $R_1 \star R_2$

Table 8.1 Operations of Relational Algebra

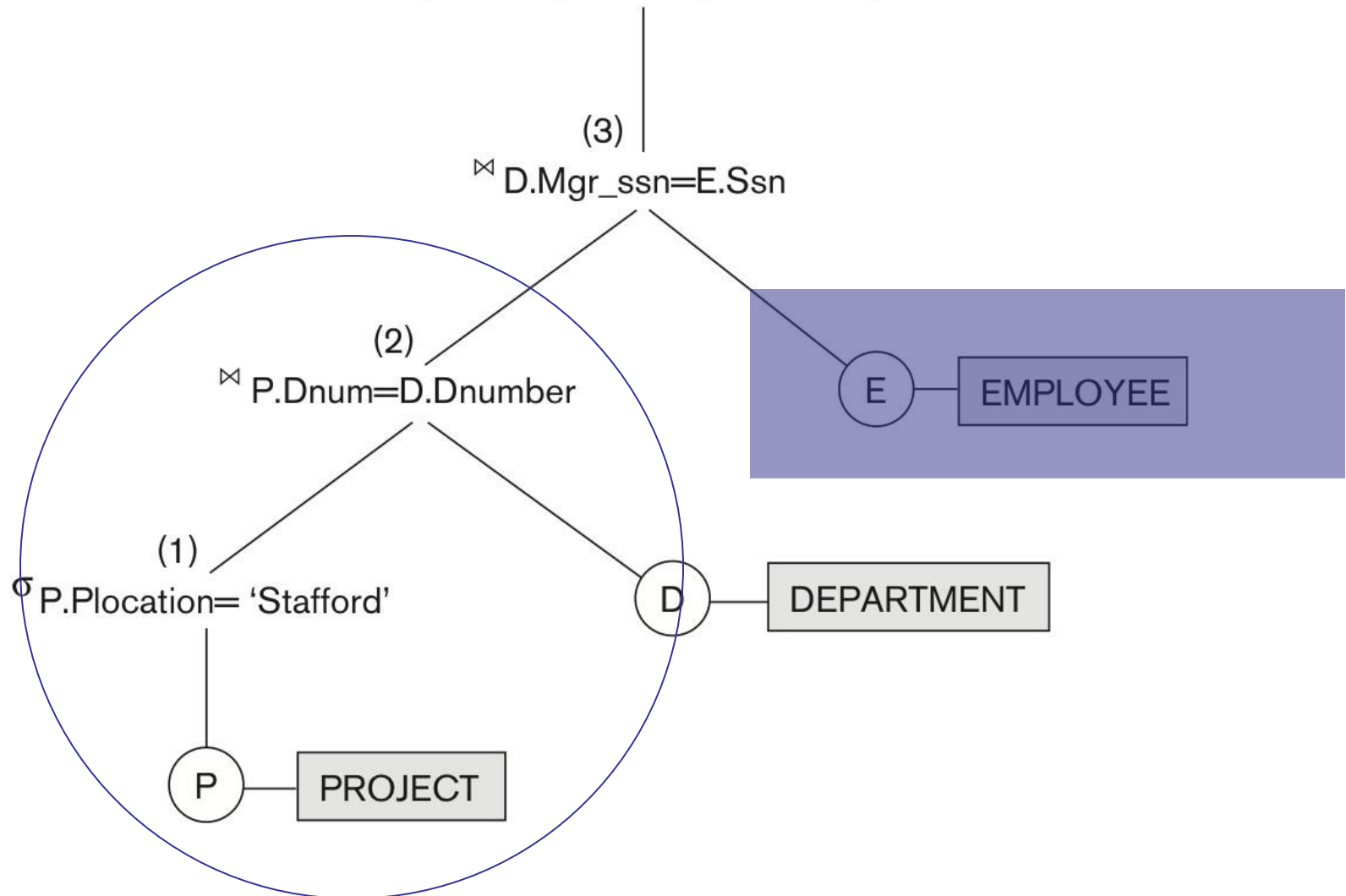
OPERATION	PURPOSE	NOTATION
UNION	Produces a relation that includes all the tuples in R_1 or R_2 or both R_1 and R_2 ; R_1 and R_2 must be union compatible.	$R_1 \cup R_2$
INTERSECTION	Produces a relation that includes all the tuples in both R_1 and R_2 ; R_1 and R_2 must be union compatible.	$R_1 \cap R_2$
DIFFERENCE	Produces a relation that includes all the tuples in R_1 that are not in R_2 ; R_1 and R_2 must be union compatible.	$R_1 - R_2$
CARTESIAN PRODUCT	Produces a relation that has the attributes of R_1 and R_2 and includes as tuples all possible combinations of tuples from R_1 and R_2 .	$R_1 \times R_2$
DIVISION	Produces a relation $R(X)$ that includes all tuples $t[X]$ in $R_1(Z)$ that appear in R_1 in combination with every tuple from $R_2(Y)$, where $Z = X \cup Y$.	$R_1(Z) \div R_2(Y)$

Query Trees

- An internal data structure to represent a query (i.e. relational algebra expressions)
- **Nodes** stand for operations like selection, projection, join, renaming, division,
- **Leaf nodes** represent ??
- Standard technique for
 - estimating the work involved in executing the query
 - the generation of intermediate results
 - and the optimization of execution

For every project located in 'Stafford', list the project number, the controlling department number, and the department manager's last name, address, and birth date.

π P.Pnumber,P.Dnum,E.Lname,E.Address,E.Bdate





To enhance the expressive power of RA:

ADDITIONAL RELATIONAL OPERATIONS

Generalized Projection

- This operation extends PROJECTION by allowing *functions of attributes* to be included in the projection list
- Notation: $\pi_{F1, F2, \dots, Fn}(R)$
- The functions over the attributes in R may involve arithmetic operations and constant values
- Example) EMPLOYEE (Ssn, Salary, Deduction, Years_service)
 Net Salary = Salary – Deduction,
 Bonus = 2000 * Years_service, and
 Tax = 0.25 * Salary

$$REPORT \leftarrow \rho_{(Ssn, Net_salary, Bonus, Tax)}(\pi_{Ssn, Salary - Deduction, 2000 * Years_service, 0.25 * Salary}(EMPLOYEE))$$

Aggregate Function and Grouping

- Common functions applied to collections of numeric values include
 - SUM
 - AVERAGE
 - MAXIMUM and MINIMUM
 - COUNT
 - used for counting tuples or values
- Grouping can be combined with Aggregate Functions
 - Grouping attributes placed to left of symbol
 - Aggregate functions to right of symbol
 - $\text{grouping_attributes } \mathcal{F}_{F1, F2, \dots, F_n}(R)$
- Example:
 - For each department, retrieve the DNO, COUNT SSN, and AVERAGE SALARY

Examples

- EMPLOYEE (SSN, Name, Salary, Year)
- $\mathcal{F}_{\text{MAX Salary}}$ (EMPLOYEE)
 - retrieves the maximum salary value from the EMPLOYEE relation
- $\mathcal{F}_{\text{SUM Salary}}$ (EMPLOYEE)
 - retrieves the sum of the Salary from the EMPLOYEE relation
- $\mathcal{F}_{\text{COUNT SSN, AVERAGE Salary}}$ (EMPLOYEE)
 - computes the count (number) of employees and their average salary
 - Note: count just counts the number of rows, without removing duplicates
- Year $\mathcal{F}_{\text{MIN Salary, AVG Salary}}$ (EMPLOYEE)
 - retrieves the minimum and average Salary values from the EMPLOYEE relation for each year.

Aggregate Function and Grouping

Report the total number of students for each year

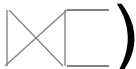
$\text{year} \mathcal{F} \text{Count sid (R)}$

S	<u>sid</u>	sname	year		year	Count_sid
	240	white	3	}	3	2
	202	jones	3			
	450	adams	1	}	1	1

Outer Join

- **INNER** joins: In THETA JOIN, NATURAL JOIN and EQUIJOIN, tuples without a *matching* tuple are eliminated from the join result
 - Tuples with null in the join attributes are also eliminated
 - This amounts to loss of information
- **OUTER** joins: can be used when we want to keep all the tuples in R, or all those in S, or all those in both relations in the result of the join, regardless of whether or not they have matching tuples in the other relation.

Outer Join

- Left outer join ()
 - Keep every tuple in the left relation in the result
- Right outer join ()
 - Keep every tuple in the right relation in the result
- Full outer join ()
 - Keep all tuples in the left and right relations in the result

Example Relations

R1

<u>A</u>	<u>B</u>	<u>C</u>
20	100	X
10	300	Y
30	400	Z

R2

<u>B</u>	<u>D</u>	E
100	1	XX
200	3	YY
100	2	ZZ

 $R1 \bowtie R2$

A	B	C	D	E

Any matches for
the 2nd and 3rd
tuple in R1?

Example Relations

R1

<u>A</u>	<u>B</u>	<u>C</u>
20	100	X
10	300	Y
30	400	Z

R2

<u>B</u>	<u>D</u>	E
100	1	XX
200	3	YY
100	2	ZZ

R1 ⋈ R2

A	B	C	D	E
20	100	X	1	XX
20	100	X	2	ZZ
10	300	Y	NULL	NULL
30	400	Z	NULL	NULL

Example Relations

R1

<u>A</u>	<u>B</u>	<u>C</u>
20	100	X
10	300	Y
30	400	Z

R2

<u>B</u>	<u>D</u>	E
100	1	XX
200	3	YY
100	2	ZZ

R1 ⋈ R2

A	B	C	D	E
20	100	X	1	XX
20	100	X	2	ZZ
NULL	200	NULL	3	YY

Example Relations

R1

<u>A</u>	<u>B</u>	<u>C</u>
20	100	X
10	300	Y
30	400	Z

R2

<u>B</u>	<u>D</u>	E
100	1	XX
200	3	YY
100	2	ZZ

 $R1 \bowtie R2$

A	B	C	D	E
20	100	X	1	XX
20	100	X	2	ZZ
10	300	Y	NULL	NULL
30	400	Z	NULL	NULL
NULL	200	NULL	3	YY

Equivalence Among Relational Algebra Expressions

- We say that expressions E1 and E2 are *equivalent* if, regardless of the tuples in the relations mentioned in E1 and E2, these expressions always return the same answers
- Simple example: Consider the relation R(A,B,C). The following three expressions are equivalent

$$\pi_A R$$

$$\pi_A \pi_{A,B} R$$

$$\pi_A \pi_{A,C} R$$

Relational Algebra Overview

- Relational Algebra consists of several groups of operations
 - Unary Relational Operations
 - SELECT (symbol: σ (sigma))
 - PROJECT (symbol: π (pi))
 - RENAME (symbol: ρ (rho))
 - Relational Algebra Operations From Set Theory
 - UNION ($\cup$), INTERSECTION ($\cap$), DIFFERENCE (or MINUS, $-$)
 - CARTESIAN PRODUCT ($\times$)
 - Binary Relational Operations
 - JOIN (several variations of JOIN exist)
 - DIVISION
 - Additional Relational Operations
 - OUTER JOINS, OUTER UNION
 - AGGREGATE FUNCTIONS (These compute summary of information: for example, SUM, COUNT, AVG, MIN, MAX)

Chapter Summary

- Relational Algebra
 - Unary Relational Operations
 - Relational Algebra Operations From Set Theory
 - Binary Relational Operations
 - Additional Relational Operations
 - Examples of Queries in Relational Algebra